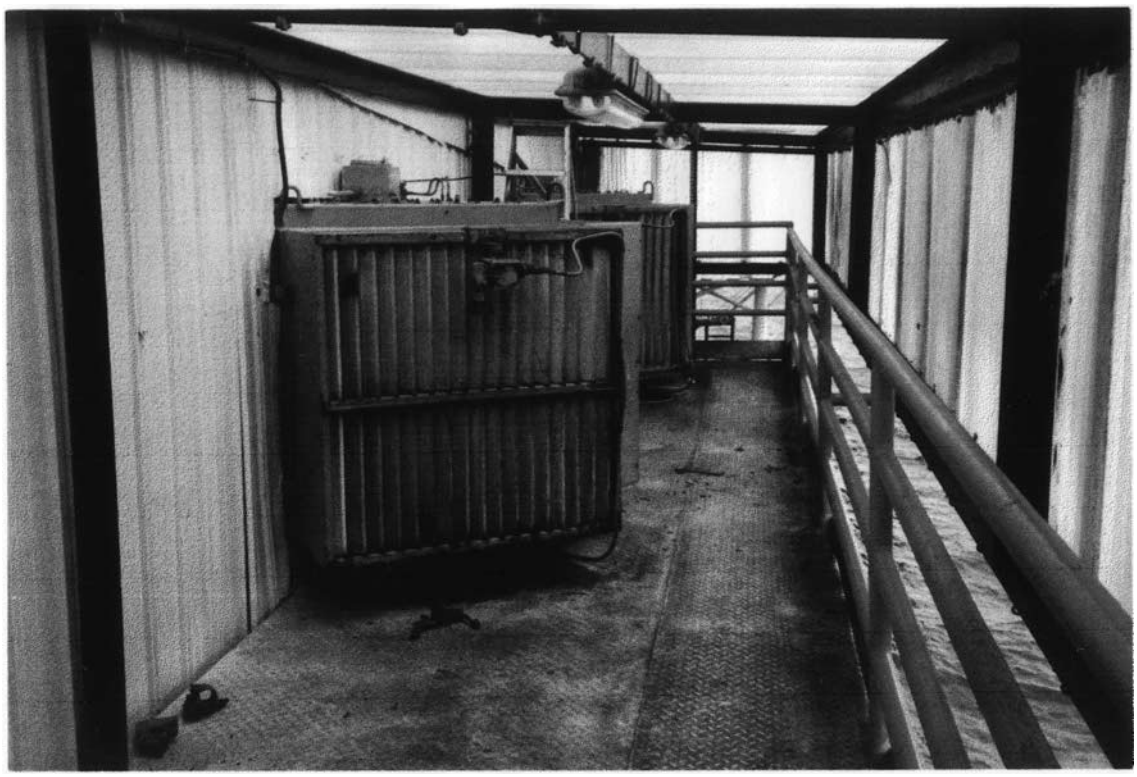


FIG. A-48 Filters protected by a canopy. (Source: Altair Filters International Limited.)

blasting. The lifetime maintenance cost of a painted carbon-steel structure vastly outweighs the small extra premium that stainless steel affords for the initial installation.

Newer Filtration Designs

Clearly, the most worrisome deficiency of the old systems was the combination of poor water-handling ability either from rain or salt-related droplets and the materials of construction.

Gas turbines that operate on warships also experience intermittent heavy-water loadings. These are protected by vane separators, which are, in essence, a high-efficiency weather louvre. The main differences are the close vane pitching, which is 3 to 4 times closer than for weather louvres, the more intricate profile of the vanes, and the integral drain system. These vanes have a high capacity and efficiency and can easily cope with even the record rainfall rates mentioned previously without the need for a weatherhood, thereby saving weight and expense. Since droplets can develop on the filter stages due to humidity, a vane separator is also necessary as the final stage to prevent the problems shown in Fig. A-51.

In addition, technology has improved, so that new, tougher filter materials are now available that will allow filters to work at slightly higher velocities and so reduce air-filter housing sizes with the associated reductions in capital and shipping costs. These tough materials can withstand shot and grit blast without damage. The higher velocities allow the systems to be designed such that droplets will break off the filter stages but then be caught in the final-stage vane separator.



FIG. A-49 Drain valve. (Source: Altair Filters International Limited.)

It is almost universally agreed that stainless steel is the most cost-effective long-term solution for construction of the air-inlet system. The premium between stainless steel over painted carbon steel can now be as low as 20%, whereas there are no further painting costs and the life is infinitely longer.

The grade of stainless steel is also important. It is recognized that the lower grades, such as American Iron and Steel Institute AISI 304 and AISI 321, do not have sufficient corrosion protection, particularly if the material is work hardened. AISI 316 is the most popular choice since it has up to 18.5% chromium, a metal whose presence helps to build up a passive protective film of oxide and prevents corrosion. Together with 10 to 14% nickel content, the steel has an austenitic structure that is very ductile and easily welded.

It also can have a low carbon content (below 0.03%) as well as a molybdenum content of between 2.0 and 3%, which increases its resistance against pitting. Indeed, one operator has paid a significant premium in both cost and delivery time by insisting that the molybdenum content be no lower than 2.5%.

Not only are the filter housings now constructed in AISI 316 but also almost all of the items such as vane separators, door locks, hinges, and instrumentation are supplied in this same material. These inlet systems will give a long life, and they look good as well. A typical system is shown in Figs. A-64 and A-65.



FIG. A-50 Salt penetration through filters. (Source: Altair Filters International Limited.)

The attention to detail is now evident. Figure A-65 clearly shows the elaborate drain systems that are now installed. In addition, the stainless-steel housings are carefully segregated in the manufacturing shop to prevent any cross-contamination from any other ferrous materials, which includes tooling.

Figure A-66 shows a Brunei 4 platform where five of the engines had been retrofitted with this system.

In summary the main requirements of a filtration system in a tropical environment are

1. Protection against tropical rainstorms by vane separators
2. The inclusion of an integrated drain system
3. The selection of AISI 316 stainless steel as the material of construction
4. Protection against droplet carryover by a final stage vane separator



FIG. A-51 Salt penetration through filters. (Source: Altair Filters International Limited.)



FIG. A-52 Corrosion at silencer outlet. (Source: Altair Filters International Limited.)

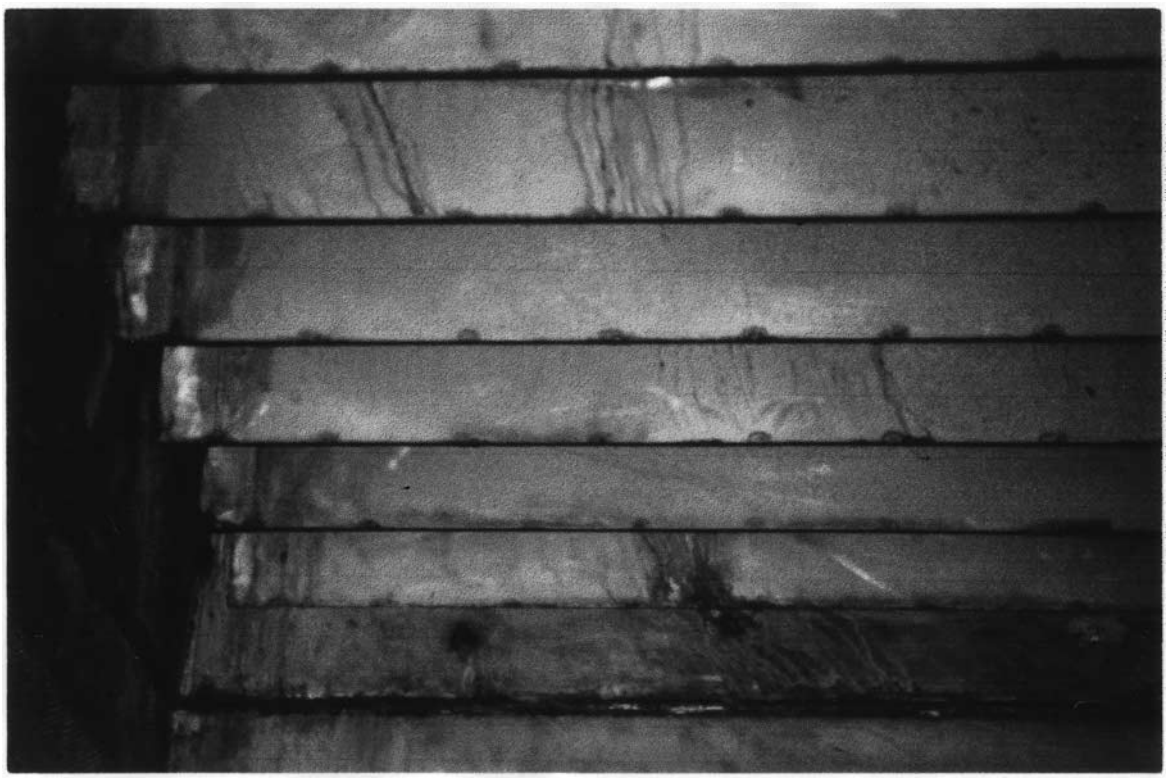


FIG. A-53 Water penetration through the inlet silencer. (Source: Altair Filters International Limited.)



FIG. A-54 Corrosion in plenum. (Source: Altair Filters International Limited.)

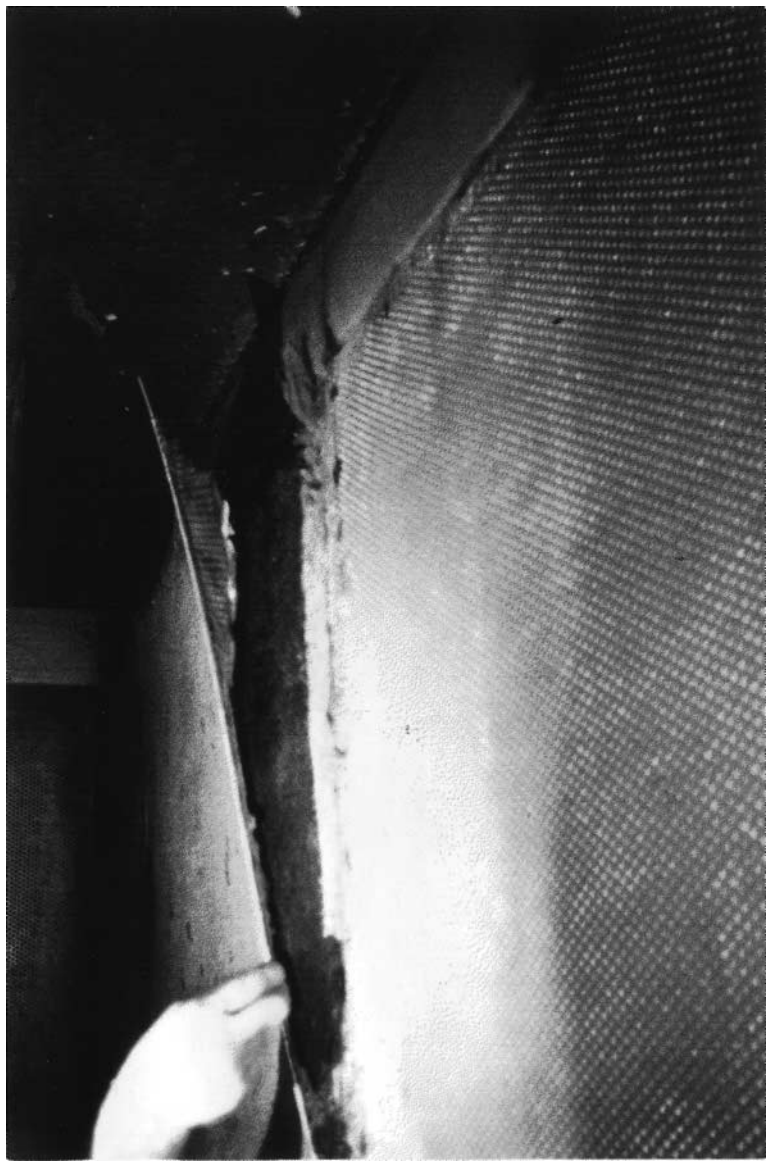


FIG. A-55 A detached plenum lining. (Source: Altair Filters International Limited.)

5. Protection against insects with an insect screen
6. The use of dust extract systems only where essential

The Offshore Environment*

In Europe in the late 1960s, the only data generally available on the marine environment was generated from that found on ships. Since at that time there was considerable interest in using gas turbines as warship propulsion systems, several attempts were made to define the environment at sea, with particular respect to warships.

*Source: Altair Filters International Limited, UK. Adapted with permission.



FIG. A-56 A corroded inertia filter. (Source: Altair Filters International Limited.)

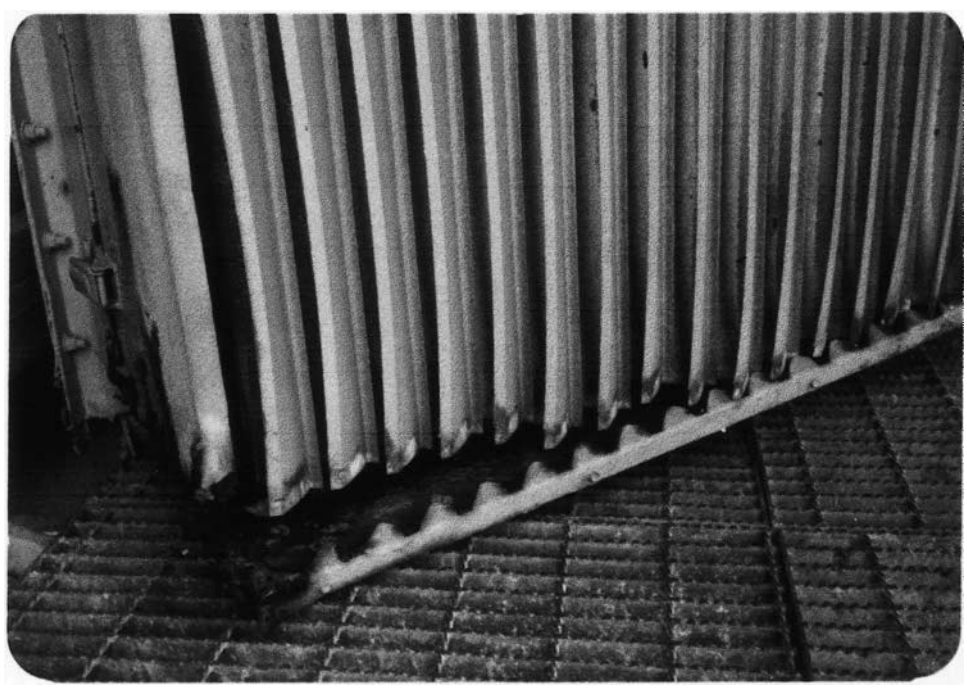


FIG. A-57 A corroded weather louvre. (Source: Altair Filters International Limited.)

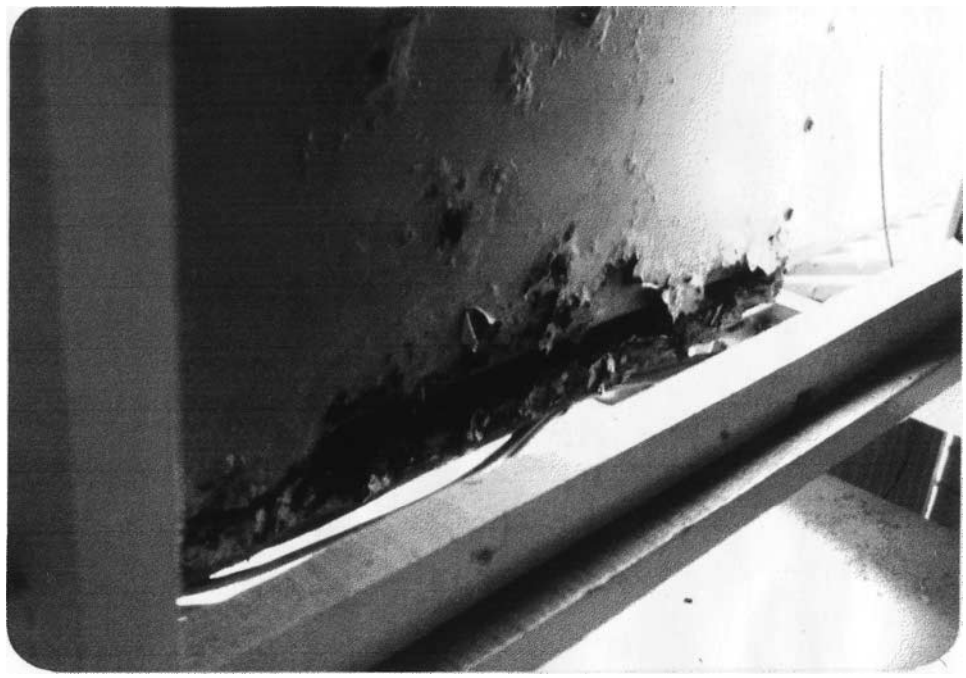


FIG. A-58 Corrosion downstream of filters. (Source: Altair Filters International Limited.)

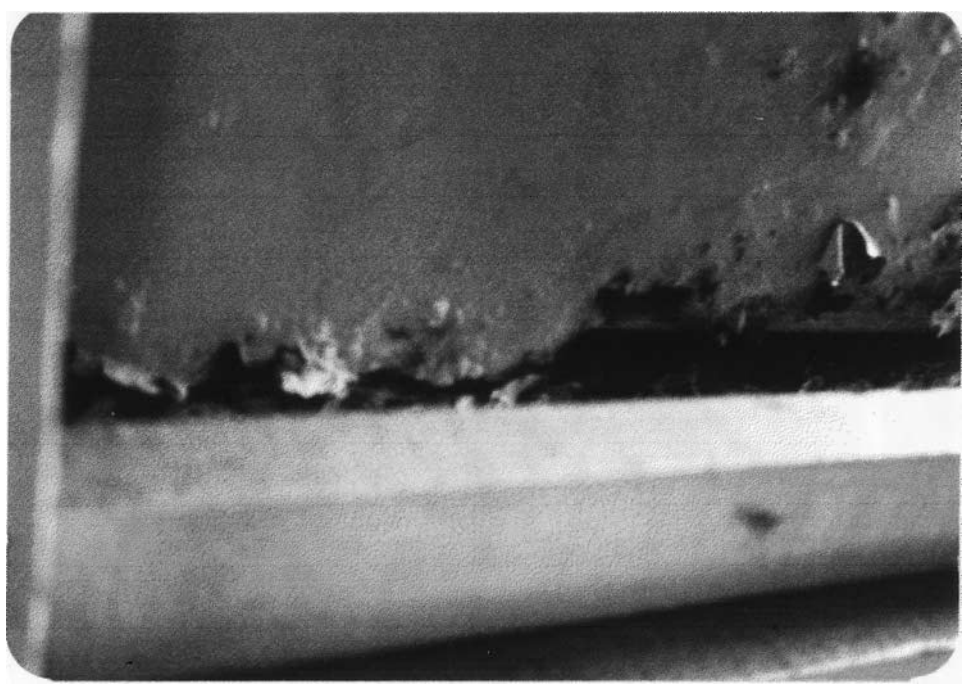


FIG. A-59 Corrosion downstream of filters. (Source: Altair Filters International Limited.)

Not only was it found difficult to produce consistent data, but other factors such as ship speed, hull design, and height above water level had major effects. It became apparent that predicting salt in air levels was as difficult as predicting weather itself.

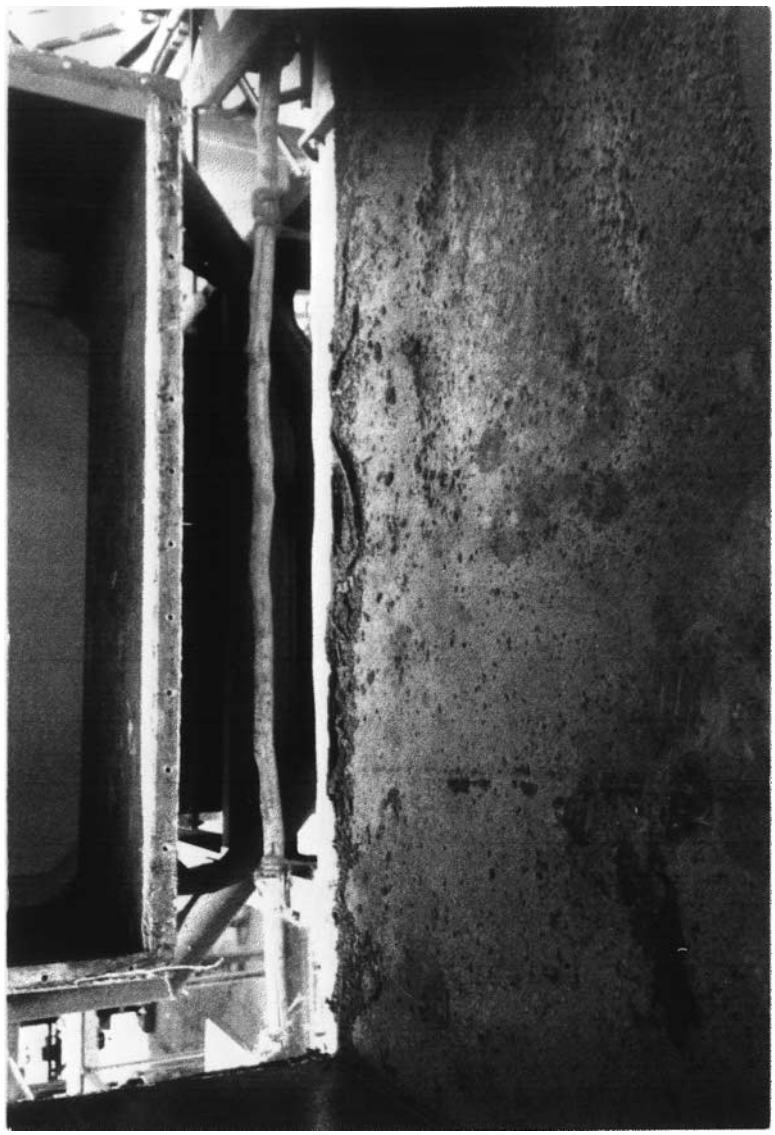


FIG. A-60 Corrosion debris in inlet duct. (Source: Altair Filters International Limited.)

Since the gas turbine manufacturers had defined a total limit of the amount of the contaminants that the turbines could tolerate, some definition of the environment was essential to design filter systems that could meet these limits.

Many papers and conferences were held with little agreement, as can be seen in Fig. A-67. However since the gas turbine industry is a conservative one, it adopted the most pessimistic values as its standard, namely the National Gas Turbine Establishment (NGTE) 30-knot aerosol (Table A-11). It was treated more as a test standard rather than what its name implied. In the absence of any other data, this was used to define the environment on offshore platforms, despite the fact that they were much higher out of the water, and did not move around at 40 knots!

This then defined the salt in air concentration, but did not address any other particulates. In hindsight, it now seems naive that the offshore environments were originally considered to be clean with no other significant problems than salt. Many



FIG. A-61 A paint blister. (Source: Altair Filters International Limited.)



FIG. A-62 A new filter housing awaiting installation. (Source: Altair Filters International Limited.)

equipment specifications were written at that time saying the environment was "dust free."

In the early 1970s there was also a lively debate as to whether the salt in the air was wet or dry. One argument was put forward that if the salt was wet it would



FIG. A-63 Corrosion on the new housing shown in FIG. A-62. (Source: Altair Filters International Limited.)

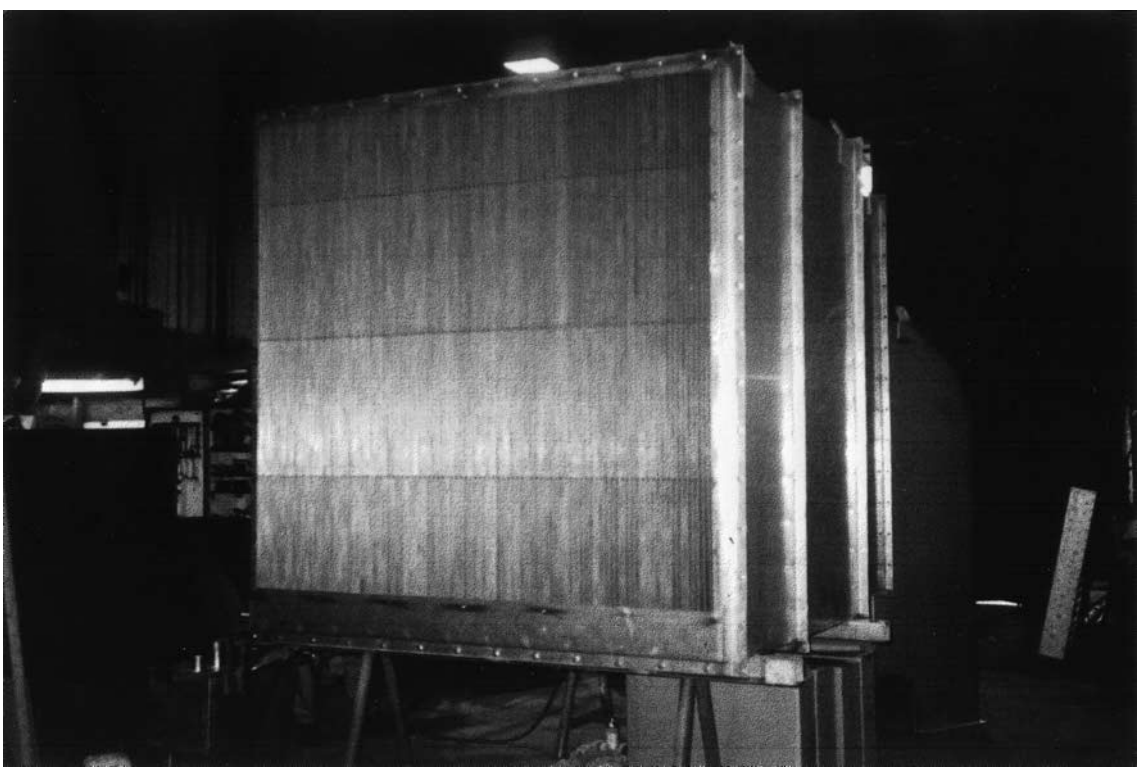


FIG. A-64 A stainless-steel filter housing. (Source: Altair Filters International Limited.)

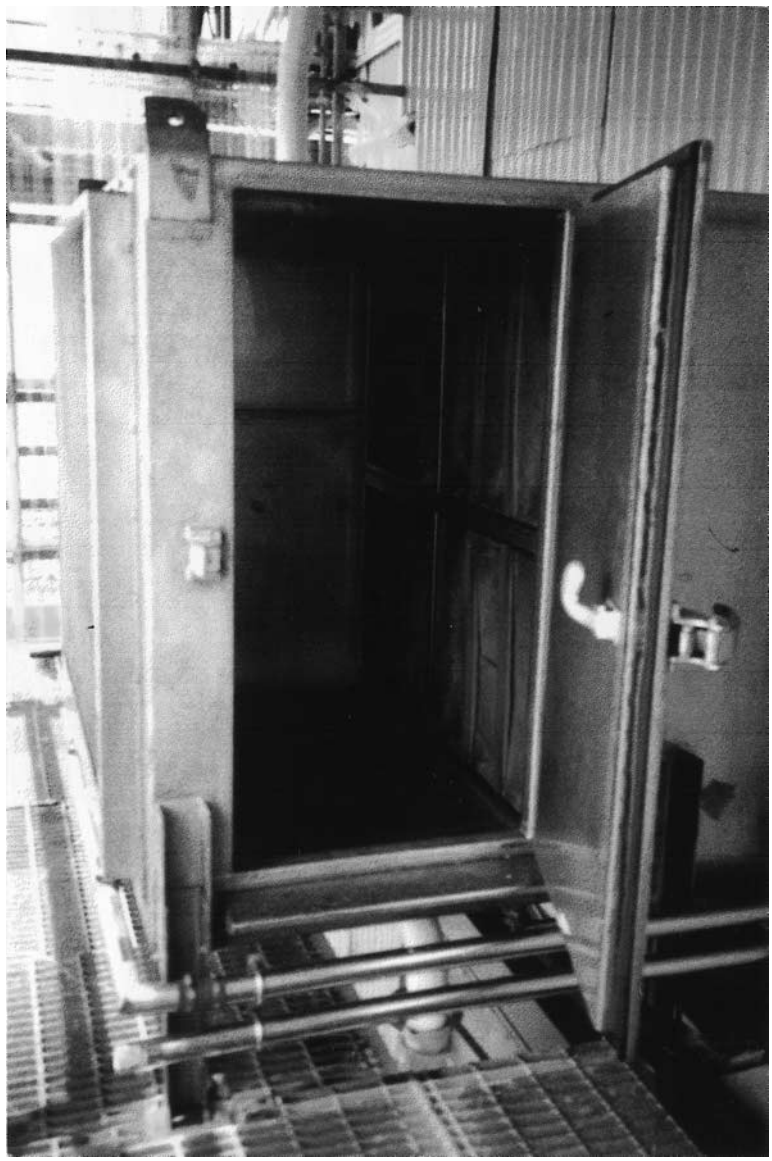


FIG. A-65 A stainless-steel filter housing. (Source: Altair Filters International Limited.)

TABLE A-11 NGTE 30-knot Aerosol

Microns	Salt Content, ppm
<2	0.0038
2–4	0.0212
4–6	0.1404
6–8	0.3060
8–10	0.4320
10–13	0.6480
>13	2.0486
Total	3.6000



FIG. A-66 Brunei shell petroleum Fairley 4 platform, showing five new filter housings. (Source: Altair Filters International Limited.)

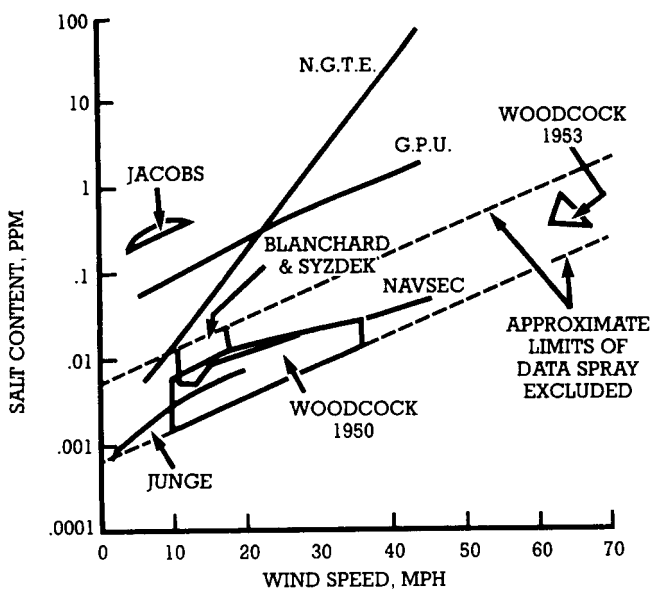


FIG. A-67 Airborne salt comparisons. (Source: Altair Filters International Limited.)

require a further stage of vane separators as the final stage to prevent droplet reentrainment from the filters. The opposing argument maintained that vane separators were unnecessary and that a lower humidity resulted in evaporation of the droplet, giving a smaller salt particle that required a higher degree of filtration. Snow and insect swarms were largely ignored as a problem.

The Initial Filter Designs

The types of systems that were used on the first phase of the developments in the North Sea fell into two categories: high-velocity systems where the design face velocity was normally around 6 m/s, and lower velocity systems that operated at 2.5 m/s.

The low-velocity system was a similar system used on land-based installations, and usually comprised a weather louvre, followed by a prefilter and a high-efficiency bag or cartridge filter. Sometimes a demister stage was added and occasionally, bleed extract inertials were supplied as a first stage.

The high-velocity system was attractive to packagers since it was lighter and occupied a much smaller space. It was the system derived from shipboard use and comprised a vane, coalescer, and vane system.

In general, both systems were housed in mild steel housings with a variety of paint finishes. The weather louvres and vanes were normally constructed from a marine grade aluminum alloy. The filter elements often had stainless steel or galvanized frames.

Bypass doors were used to protect the engine against filter blockage.

The emphasis by package designers was to include a provision for a "3 or 4 stage system" often without regard for what those stages should comprise.

The Actual Offshore Environment

The actual offshore environment is in many ways different from that originally envisaged.

Salt in air is present, although it is only a problem when the filtration system leaks or is poorly designed. Horizontal rain can be a severe problem although sea spray does not generally reach the deck levels even in severe storms.

Flare carbon and mud burning can be a significant problem if the flare stack is badly positioned or if the wind changes direction. (See Fig. A-68.) Not only do the filters block more quickly, but greasy deposits can cover the entire filter system, making the washing of cleanable filters more difficult.

The relative humidity offshore was found to be almost always high enough to ensure that salt was in its wet form. Some splendid work by Tatge, Gordon, and Conkey concluded that salt would stay as supersaturated droplets unless the relative humidity dropped below 45%. Further analysis of offshore humidities in the North Sea showed that this is unlikely to happen (Table A-12).

Initially it was thought that the platforms were dust free, but this is far from the case.

Drilling cement, barytes, and many other dusts are blown around the rig as they are used or moved. But the main problem has resulted from grit blasting. As the platforms got older, repainting was found to be an accelerating requirement with grit blasting a necessary prerequisite.

TABLE A-12 Monthly Average Relative Humidity, North Sea

January	91	July	85
February	86	August	83
March	87	September	83
April	84	October	80
May	86	November	81
June	84	December	80

Source: ASME Report 80-GT-174.

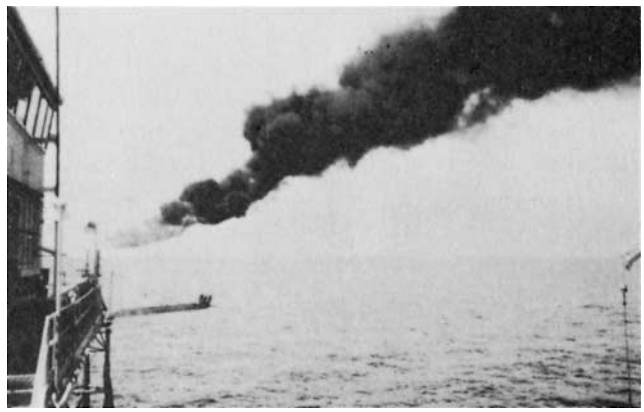


FIG. A-68 Flare carbon can cause problems. (Source: Altair Filters International Limited.)

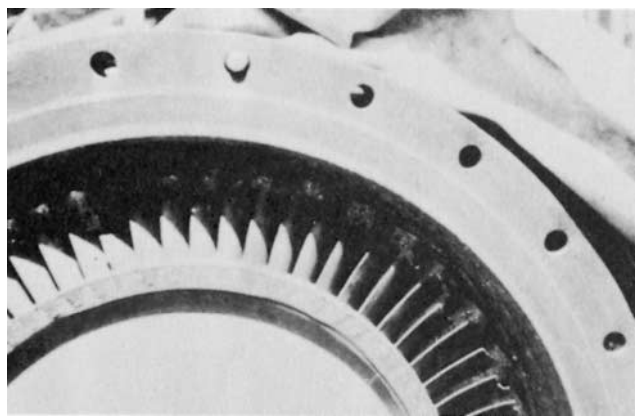


FIG. A-69 Typical turbine damage. (Source: Altair Filters International Limited.)

In order to be effective, grit is sharp and abrasive by design and can be devastating if ingested into a gas turbine. (See Fig. A-69.) The quantities used can seem enormous. On one platform it was found that over a 12-month period, 700 tons of grit blast had been used!

The Problems Encountered

In general, problems were slow to appear, typically taking three to five years after start-up, but since a lot of equipment had been installed at about the same time, the problems manifested themselves like an epidemic.

These problems could be categorized as follows:

1. Erosion of compressor blading
2. Short intervals between compressor cleaning
3. Frequent filter change-out
4. Turbine corrosion
5. Corrosion of the filters and housing

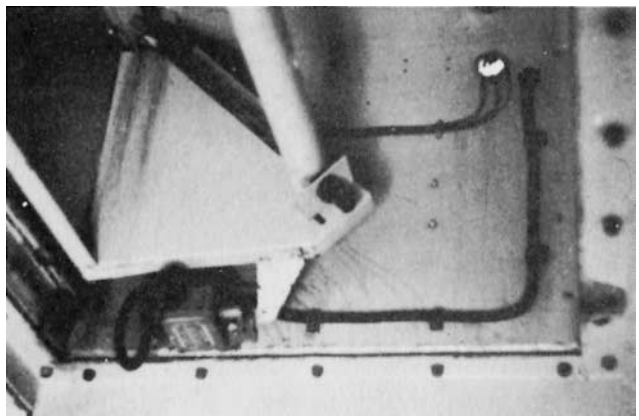


FIG. A-70 Typical leak caused by a missing cable gland. (Source: Altair Filters International Limited.)

By far the most serious of these problems was the erosion of compressor blading that was experienced almost simultaneously on many platforms. This occurred about three to five years after start-up, as this was the time that repainting programs were initiated. Grit blast found its way into the turbine intakes either through leaking intakes, bypass doors, or through the media itself. (See Fig. A-70.) Since the airborne levels were high, the air filters quickly blocked up, allowing the bypass doors to open. As filter maintenance is not a high priority on production platforms, considerable periods were spent with grit passing straight into the turbine through open bypass doors. Even where maintenance standards were more attentive, there were usually enough leaks in the intake housing and ducting to ensure delivery of the grit to the turbine.

It often seemed contradictory that the system designers would spend a lot of time specifying the filter system, but would pay little attention to ensuring the airtightness of the ducting downstream.

Since the grit was sharp, it sometimes damaged the filter media itself, reducing the system efficiency dramatically.

Bypass doors were a major problem. Early designs failed to take account of the environment or the movement in the large structures of the filter housings. Very few of those initial designs were airtight when shut, and it was not uncommon for them to be blown open by the wind.

Turbine corrosion could almost always be traced to leaky ducting or operation with open bypass doors. Very few systems gave turbine corrosion problems if the ducting was airtight. The few installations that did give problems were usually the result of low-velocity systems operating with poor aerodynamics, so that local high velocities reentrained salt water droplets into the airstream and onward to the engine. (See Fig. A-71.)

Rapid compressor fouling was usually the prelude to more serious problems later, since it was usually caused by the combined problems of filter bypass.

Compressor cleaning almost once a week was fairly standard for systems with those problems.

As time progressed the marine environment took its toll on the carbon steel and severe corrosion was experienced on the intake housing and ducting. (See Fig. A-72.) In some cases, corrosion debris was ingested into the turbine causing turbine failure. This again was accelerated by poor design, which allowed dissimilar metals to be put into contact, leading to galvanic corrosion.

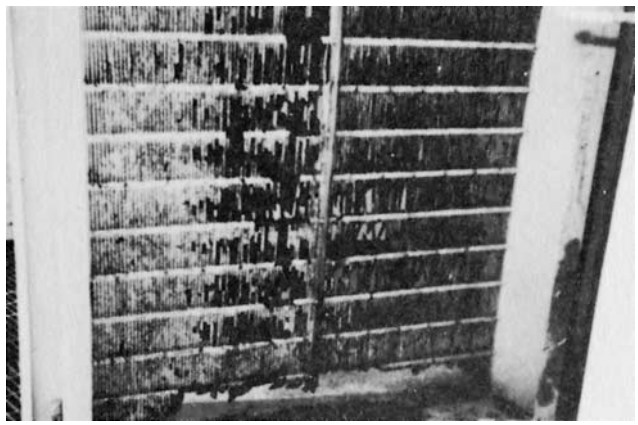


FIG. A-71 Inertial filters showing severe corrosion. (Source: Altair Filters International Limited.)

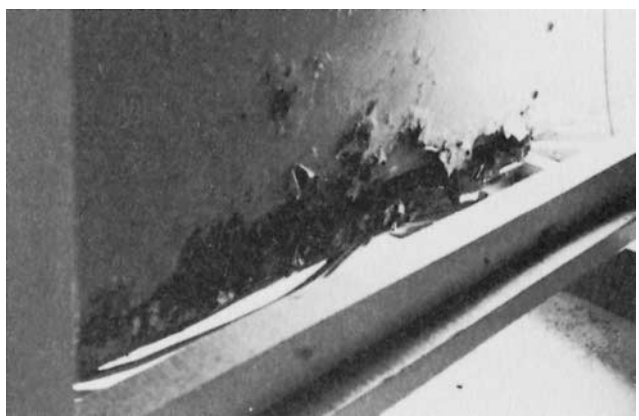


FIG. A-72 Severe duct corrosion. (Source: Altair Filters International Limited.)

Specification of a Typical Filter Used in the Offshore Environment*

Gas turbines are an ideal power source for driving compressors, pumps, and generators. Since they are relatively small compared to their power output, they can be used easily in remote locations such as jungles, deserts, and offshore platforms. They are, however, very complex pieces of machinery, comprised of many high-tolerance rotating parts.

The engineering is further complicated by the engine manufacturers' need to increase the turbine efficiency by increasing operating temperatures. In order to overcome the material stresses associated with these higher temperatures, internal cooling passages have been introduced into the engine. Typically, turbine blades are now of hollow construction with cooling air blown through them, exiting through tiny holes in the blade surface. These holes can be very small and are very susceptible to blockage. The requirement for filtration of the gas turbine air is, therefore, even more stringent than in the past. The need to filter the air to the gas turbine is fourfold:

* Source: Altair Filters International Limited, UK. Adapted with permission.

- To prevent erosion
- To protect against fouling
- To prevent particle fusion
- To protect against corrosion

Erosion

Erosion is caused by particles impacting and wearing away the metal surfaces. The high speeds of the rotating blades collide with the airborne particles and produce a quite large change in energy, which results in fragments of metal being blasted out of the blade surface. Even particles as small as 10 μm in diameter can cause severe erosion. The composition and shape of the particle can also significantly affect the erosion rate. Blade profiles are so carefully designed that even minor abrasions can alter the profiles to such an extent that engine performance is affected. Erosion is an expensive problem since it causes permanent damage and the affected parts will require replacing. It is proportional to concentration and in severe duties, such as gas turbine–powered hovercraft operated in desert conditions, engine life has been reduced to as little as 6 hours.

Fouling

Engine fouling, by comparison, is normally only a temporary problem and is caused by a buildup of contamination that adheres to the internal surfaces.

Again, deposition on blade surfaces can change profiles, with the resultant loss in engine power and an increase in fuel consumption. Particles of 2 μm and less are generally the major cause of fouling. Smoke, oil mist, and sea salt are common examples.

The particles are attracted to the metal surfaces by a variety of forces, including impaction, electrostatic, and capillary action. The composition of the particle, again, is important in determining the rate of fouling. In marine environments, dry dust particles are often coated in a layer of sea salt, which is viscous by nature and adds to the fouling action. While fouling is basically a temporary problem, it can be removed by various cleaning techniques. It is an irritant to the operator, as many of these cleaning processes have to be conducted at reduced powers or with the engine stopped. In the past, engines were cleaned by injecting a mild abrasive into the engine to clean off the contamination when the engine was running. While the most common material was a mixture of ground coconut shells, rice has been used on some tropical engines. The practice of using online cleaning has now mostly been abandoned since it tended to transfer large particles of debris into other areas of the turbine, causing even more problems. There was also a view put forward that it accelerated hot end corrosion. Modern cleaning methods use a detergent sprayed into the engine on a cold cycle, leaving it to soak and then washing it off with clean water.

Particle fusion

Dry particles, which range in size from 2 to 10 μm, could, on old engine designs, pass through the engine causing little or no damage. However, on the new generation of hotter engines, these particles can cause problems if their fusion temperature is lower than the turbine operating temperature, since they will melt and stick to the hot-metal surfaces. This can cause severe problems since this molten mass can block cooling passages and cause thermal fatigue. The affected surface is permanently disfigured and will need replacement.

Corrosion

The high temperatures of the gas turbine can also cause rapid acceleration of the corrosion process. Even though the hot-metal surfaces are made of some of the most sophisticated materials, corrosion can still be extremely rapid. Blade failures in as little as 100 operating hours have been known, and failures within 2000 operating hours are relatively common. Corrosion, however, can be completely prevented by modern techniques, and yet it still occurs.

Normally, corrosion is produced by a salt, such as sodium or potassium, but lead and vanadium are also common contributors. Since many gas turbines are based either offshore or close to the sea, sea salt (sodium chloride) is the main offender. In the cold parts of the engine it is the sodium chloride that does the damage, whereas in the hot parts of the engine it is sodium sulfate (or sulfidization) that creates most of the corrosion. Sodium sulfate is produced from the combination of sulfur in the fuel and sodium chloride in the air.

It is important to recognize that the corrosion process is self-propagating, and, once started, will continue even though the source problem has been cured.

The modern gas turbine therefore is a sensitive machine and needs to be protected to provide an acceptable life cycle. For this reason, there are limits that are recommended by the manufacturers in order to achieve this. There is not one universal limit that is adopted by all manufacturers. Each has its own, which is expressed in many different forms, either as an absolute limit or one that is time dependent. However, all seem to work from the same premise.

Previously, it was often thought that providing a gas-turbine air-filter system was well chosen; it could be used in almost any environment with equal effect. This has proved to be a fallacy, as many operators have found at their cost.

The Problems Solved

Since engines appeared to be eroding at a fairly rapid rate, irrespective of which type of filter system was fitted, an equally rapid response was required.

Phillips decided, after removing a GT22 engine from their Ekofisk Bravo platform, that they would not operate the repaired engine until the filter system was changed, even though the source of the problem had not been identified at that time.

An evaluation of all the likely contaminants was quickly undertaken, with large quantities of various suspicious substances shipped back to the laboratory for analysis and trial against the installed filter system in the wind tunnel. Grit blast was confirmed as the source of erosion, and the installed filter system gave only a 28% protection.

Trials against other conventional filter medias proved negative, since the grit was sharp and eventually cut its way through the media.

A new type of filter was required with a very strong media and a large capacity to absorb the huge quantities of grit without blocking too quickly. The ability to work at high velocity would also be an advantage.

This was achieved by using a bonded polyester fiber, which proved almost impossible to tear. The strength of the polyester allowed the media to be operated at higher velocities without fear of fiber loss, which can be a limitation with the brittle glass fibers used in conventional filtration.

The old filter makers' methods of packing in more media with the hope that the increased area would reduce the media velocity has distinct limitations since the extra area is not effectively used by the poor aerodynamics created. Instead a careful aerodynamic design ensured a more even distribution throughout the filter,

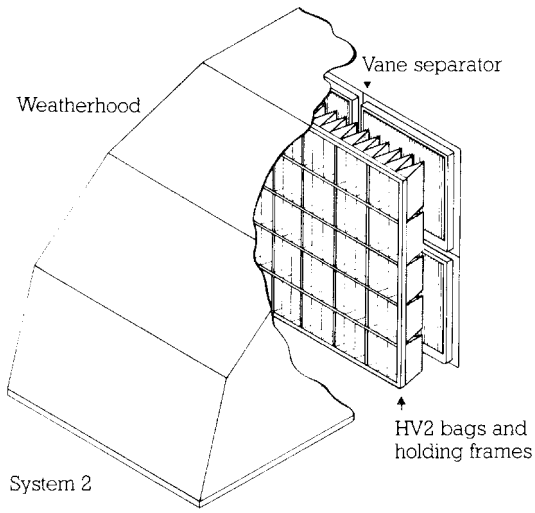


FIG. A-73 Typical air filter on a platform. (Source: Altair Filters International Limited.)

giving a relatively wide pocketed bag filter capable of those higher velocities and giving protection against the problem contaminants. By having a relatively high loft to the media, protection against filter blinding was ensured, with a resulting longer life. Testing showed the new HV2 filter to be over 99% efficient against the damaging grit blast.

A cleanable prefilter bag of similar but less dense media was also quickly designed. This PB1 prefilter bag was unique in that it was designed to be tucked inside the final filter bag and so only took a further 25-mm installation depth.

The new filters were installed into a new stainless steel housing protected front and back by vane separators and delivered on Christmas Eve 1983, just three weeks after the initial problems were investigated; a record for which all those involved have a right to be proud of. (See Fig. A-73.)

Shell was the next platform operator to experience similar problems, firstly on the Avon gas turbines on its Brent Delta platform. In these installations it was possible to fit the HV2 filters in an access space between the existing filter stages. Shell undertook a bold and very correct decision to weld up the troublesome bypass doors, having first revised the alarm and trip levels for the intake depression pressure switches. This system was carefully monitored for a period of nine months before the remaining 23 Avons on the Brent platforms were similarly converted. Comparison of two adjacent engines, one with the original system and one with a retrofitted system showed that over a three-month period the performance of the retrofitted engine was unchanged, while the other engine showed a steady increase in exhaust temperature for a given power output amounting to 30°C at the end of the period. Also, the requirement to change out the filters was reduced from 350 hours to, in some cases, over two years.

On Shell Leman BK platform, a similar comparison of two Avons was investigated, with an air sampling program constantly monitoring the quality of the inlet air over a period of one month. This showed the modified installation to be 10–14 times more effective, in terms of particle penetration.

It is not surprising that Shell has now retrofitted 78 installations worldwide. In total this design has been selected for 212 new and retrofitted gas turbine engines.

Filter systems that were an operational irritation every 15 days or so are now forgotten to such a degree that on some platforms the filters have operated, without replacement, for up to three years.

Compressor cleaning is operated on a planned maintenance approximately every 2000 hours.

The Systems of the Future

With so much experience gained with these retrofitted installations, new installations are now designed taking account of the lessons learned.

Typically a new offshore system will comprise the following features:

1. A housing made entirely of 316 stainless steel
2. A weather hood or high-efficiency weather louvre constructed in 316 stainless steel
3. A prefilter and filter stage capable of high efficiency against grit blast and other contaminants
4. A final vane separator to protect against droplet reentrainment
5. No bypass doors
6. Pressure switches for alarm and shutdown
7. An integrated drain system
8. All materials capable of withstanding a marine environment, with an exclusion of dissimilar metals, cardboard, and the like
9. A leak-free intake system

The argument for the high-velocity (6 m/s) system is now proven, with over 200 installations worldwide. The advantage of smaller size and lower weight will become more important in the future, and may push the current designs even further.

The key components of the system* are:

High-efficiency filtration. (Hercules™)

Dynamic water eliminator. High-efficiency separator ensures that salt carryover problems are eliminated (Hydra™)

Hercules and Hydra combine to form System Aquila (Fig. A-74), providing the following features:

High volume flow. Leading to a filter house with a 65% smaller face area than traditional systems. This means a customer saves space and weight, which also saves cost. (See Fig. A-75.)

High efficiency. High dust arrestance and salt removal efficiencies provide excellent protection from erosion, corrosion, and fouling of turbine blades. (See Fig A-76.)

Low pressure loss. A typical clean system pressure loss of only 45-mm H_2O means lower fuel consumption, higher output, and longer filter life for operators.

High-efficiency filtration

This feature has been aerodynamically designed to ensure that maximum particulate efficiency is achieved with the minimum resistance to airflow. The

* Note: Trademarks are specific to the source for this section. Each manufacturer will have its own equivalent terms and trademarks.

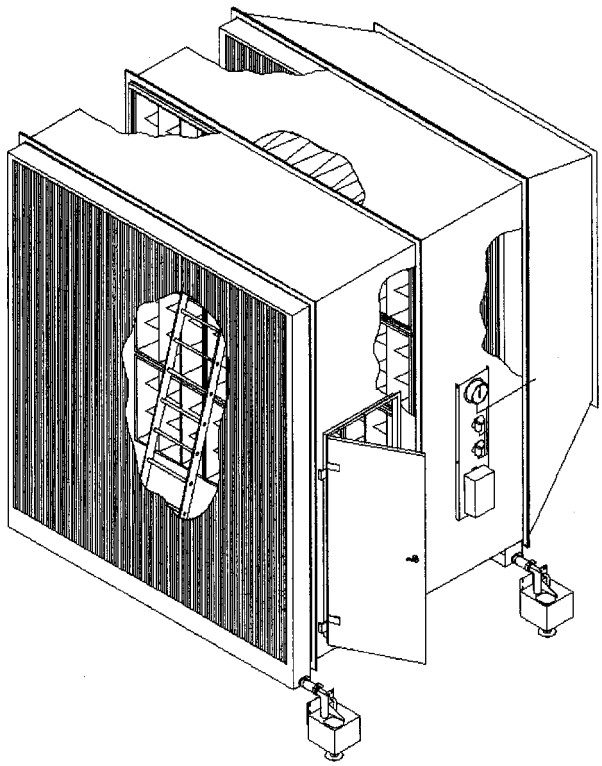


FIG. A-74 Filter with water eliminator. (Source: Altair Filters International Limited.)

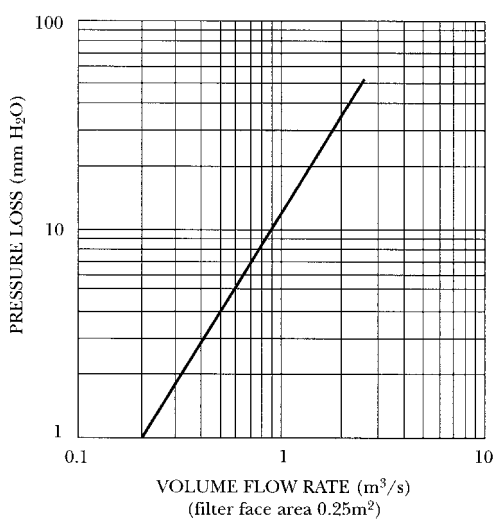


FIG. A-75 Pressure loss versus volume flow rate filter characteristic. (Source: Altair Filters International Limited.)

semirigid construction, together with the fact that each pocket is divided into smaller segments by means of a semipermeable "shelving" system, ensures the best possible profile throughout all operating conditions. This produces an extremely uniform flow distribution, leading to improved dust-holding capacity and eliminating the likelihood of localized dust breakthrough.

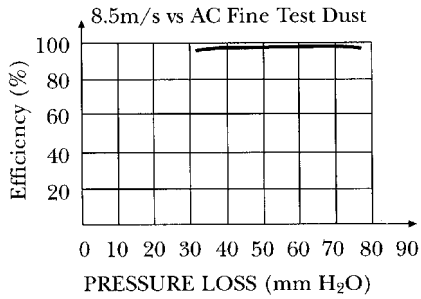


FIG. A-76 Efficiency versus pressure loss filter characteristic. (Source: Altair Filters International Limited.)

Dynamic water eliminator

This feature conducts water and salt removal. The vanes, which are constructed from corrosion-resistant marine grade aluminum (other materials are available), are produced with a profile that allows the maximum removal of salt and water, yet produces an extremely low pressure loss. This optimal profile has been achieved by the very latest design methods, and in particular by utilizing a Computational Fluid Dynamics (CFD) flow modeling system. Hydra also incorporates a unique and novel method of separating water droplets from the air stream, and this has led to improvements in bulk water removal compared with conventional methods.

Reference and Additional Reading

1. Tatge, R. B., Gordon, C. R., and Conkey, R. S., "Gas Turbine Inlet Filtration in Marine Environments," ASME Report 80-GT-174.

Typical Specifications for Range of Air Filters

This range includes panels and bags as well as high-efficiency, high-velocity systems and air/water separators.

Filter holding frames are constructed in mild or stainless steel, designed to provide quick and easy removal from upstream, downstream, or sides of ducting, without the use of springs or clips of any kind. Filter housings, ducting, louvres, dampers, and silencers can also be designed and fabricated, providing a total system capability.

Air Pollution Control*

The main methods of combating and controlling air pollution include:

- Electrostatic precipitators (for particulates)
- Fabric filters (for dust and particulates)
- Flue gas desulfurization (for SO_x removal)
- SCR $DeNO_x$ (for NO_x removal)
- Absorbers (for environmental toxins)
- End-product–handling systems (for solid and liquid wastes)
- Combined unit systems (for some or all of the previous items)

*Source: Alstom. Adapted with permission.

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